

Maksuda Aktar Toma

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Research interests

Biostatistics, Bioinformatics, Epidemiology, Machine Learning, Deep Learning, Data Visualisation and Analytics, Bayesian Analysis, Casual Inference

Education

University of Nebraska Lincoln 2025 – Present

PhD in statistics (in progress) CGPA: 3.700/4.00

University of Nebraska Lincoln 2023 – 2025

Master of Science in Statistics CGPA: 3.634/4.00

University of Dhaka 2020 – 2021

Master of Science in Statistics CGPA: 3.221/4.00

University of Dhaka 2016 – 2020

Bachelor of Science in Statistics CGPA: 3.284/4.00

Research experience

Feeding Behavior - Based Detection of Health Status in Pigs Using ML Modeling

Mentor: Dr. Sruti Das Choudhury and Dr. Tami Brown-Brandl (University of Nebraska Lincoln) Aug 2025 – Present

Applied data analytics, segmented regression, and machine learning to model feeding behavior patterns from a USDA swine production dataset. Designed algorithms to detect behavioral change points indicating disease onset and recovery, integrating LASSO, Ridge, and Random Forest models for multiclass health classification. Conducted extensive cross-validation, hyperparameter tuning, and diagnostic evaluation. [here](#).

Data-driven Techniques for Pea Genetic Diversity Analysis using Morpho-Agronomic Traits

Mentor: Dr. Sruti Das Choudhury and Dr. Dipak Santra (University of Nebraska Lincoln) Nov 2025 –Present

This study aims to characterize 329 pea germplasms using 25 key morpho-agronomic traits to assess genetic diversity and inter-trait relationships. I apply exploratory data analytics, principal component and factor analyses, and step-wise regression to identify critical traits influencing yield per plant and yield per plot, offering quantitative perspectives on pea improvement and selection.

100 Years of Pies vs. Bars

Mentor: Dr. Susan Vanderplas (University of Nebraska Lincoln) Jan 2025 – Aug 2025

This project reviews 100 years of pie vs. bar chart study on graphical perception. Pie charts are useful for estimating proportions, whereas bar charts are better for comparing data. The ideal option depends on the goal, not aesthetics or tradition. [here](#).

Academic Projects

Sentiment Analysis and Sequential Models with IMDB Data

Designed and evaluated 16 LSTM and Transformer models for IMDB sentiment classification, implementing attention, dropout, and regularization. The best LSTM model achieved 88.3% test accuracy. Validation curves, confusion matrices, and error analyses showed that LSTMs outperformed Transformers in generalization.

Convolutional Architectures Analysis on CIFAR-100

Designed and evaluated 16 CNN models on CIFAR-100, varying depth, optimizers, learning rates, dropout, and L2 regularization. A hierarchical CNN achieved 52.0% Top-1 and 82.4% Top-5 accuracy, outperforming the best baseline (50.1% / 79.5%). Results showed that deeper CNNs with Adam and tuned hyperparameters generalize better than shallow, SGD-based models

Forecasting Rideshare Prices Using Data Analytics and Machine Learning Models

Executed data preprocessing involving missing value imputation, temporal feature engineering, one-hot encoding, and outlier detection utilising the interquartile range (IQR). Five machine learning models (Linear Regression, Elastic Net, KNN, Random Forest, XGBoost) were implemented and tested for fare prediction, with XGBoost demonstrating superior performance ($R^2 = 96.16\%$) and minimal error rates.

Teaching experience

Graduate Teaching Assistant, Department of Statistics (University of Nebraska Lincoln)

Fall 2023 – Present

STAT 218: Introduction to Statistics

STAT 380: Statistics and Applications

Instructor for introductory and intermediate undergraduate statistics courses covering probability concepts, statistical inference, and regression analysis. Developed, administered, and graded assignments and exams, providing detailed feedback to strengthen student understanding.

Learning Assistant, College of Agricultural Sciences and Natural Resources (CASNR) (University of Nebraska Lincoln)

Fall

2025

	SCIL 101: Science and Decision Making - For a Complex World Mentoring 28 students across 7 project groups, guiding them through experiential learning and a semester-long showcase project. Leading active learning discussions, assess student learning, and provide feedback on assignments and final projects.	
Publications	Factors influencing non-academic achievements of Bangladeshi students - An online questionnaire-based study (Preprint) Md. Rabiul, Md. Hasibur Rahman, Maksuda Aktar Toma, Pranta Das <i>Research Square</i>, 2021. DOI: 10.21203/rs.3.rs-410311/v1	
Industry experience	Association for Social Advancement (ASA) Research Officer Conducted mixed-method analyses to evaluate program outcomes and risks, designed surveys, ensured data quality, and presented actionable insights and reports to stakeholders	Dhaka, Bangladesh Apr 2023 – Jul 2023
	Centre for Governance Studies (CGS) Research Assistant Conducted qualitative research on good governance, led focus groups for a U.S. State Department project on misinformation, and co-authored articles on societal issues	Dhaka, Bangladesh Apr 2022 – Mar 2023
	Ethical Trading Initiative (ETI) Intern Performed Data analysis, data validation, reporting, and qualitative research (FGD/KII).	Dhaka, Bangladesh Jan 2021 – Jul 2021
Talks	100 years of Pies vs. Bars 2025 International Indian Statistical Association Kansas Math Symposium 2025	
		June 2025 April 2025
Skills	Programming and Tools Proficient in: R, Python, TensorFlow, SAS, STATA, SPSS, SQL, PowerBI Familiar with: C, C++, Fortran Languages English, Bengali, Hindi	
Service and outreach	Tresurer, Statistics Graduate Student Association (GSA), UNL July 2024 – Present	

Manage the budget, coordinate funding requests, and support academic and social events in collaboration with the executive board.

General Secretary, Dhaka University Debating Society (DUDS) Oct 2021 – Dec 2022

Organised multiple national and international debate festivals, managing logistics, partnerships, and teams. I won several championships in tournaments.

Professional
memberships

American Statistical Association (ASA) July 2024 – Present

Attended AMSTAT Nebraska Travelling Course "From Big Data to Better Insights: A Primer on Machine Learning Methods in a Data-Centric World."

International Indian Statistical Association (IISA) Jun 2025 – Present

Attended the conference and took the short courses "Multimodal Causal Inference for Data Science and Biomedical Research" and "Boosting R Code performance via C++ Integration within Rstudio".